

Open-Vocabulary Detection of Fine-Grained Objects

Miro Kakkonen

Center for Machine Vision and Signal Analysis (CMVS)

University of Oulu

Oulu, Finland

miro.kakkonen@oulu.fi

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I. INTRODUCTION

General detection of arbitrary objects is a fundamental problem in computer vision. While the performance of object detection models has improved significantly in recent years, conventional models are limited to detecting a pre-defined set of object categories that they have been trained to recognize. This limits their use to specific applications where the set of objects of interest is known in advance. [1] [2]

Open-vocabulary detection (OVD) attempts to address this limitation by utilizing large-scale vision-language models (VLMs) to enable the detection of objects not based on a pre-defined set of classes, but their textual descriptions. This allows for a potentially unlimited number of object categories to be detected, including fine-grained categories that may not be well-represented in existing datasets. In this paper, we investigate the performance of state-of-the-art OVD models and VLMs on the task of fine-grained object detection for augmented reality (AR) applications in the context of guided repairs. We evaluate the models on a custom dataset of images of various objects commonly found in AR repair scenarios, and analyze their performance in terms of detection accuracy and generalization to unseen object categories.

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II. RELATED WORK

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IV. METHODOLOGY

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V. RESULTS

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VI. DISCUSSION

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VII. CONCLUSION

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